

EXPLORATORY DATA ANALYSIS

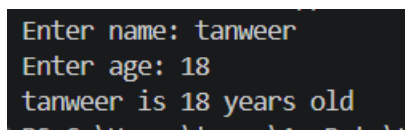
BASIC PYTHON PROBLEMS

-TANWEER M, 25BAI0201

1) INPUT AND FORMATTED STRING OUTPUT:

```
name=input("Enter name: ")
age=int(input("Enter age: "))
print(f"{name} is {age} years old")
```

OUTPUT:

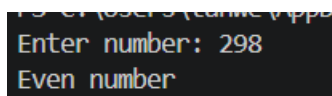


```
Enter name: tanweer
Enter age: 18
tanweer is 18 years old
```

2) EVEN OR ODD:

```
num=int(input("Enter number: "))
if num%2==0:
    print("Even number")
else:
    print("Odd number")
```

OUTPUT:

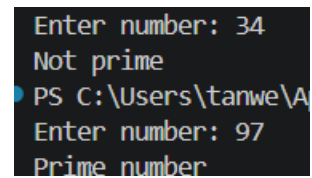


```
Enter number: 298
Even number
```

3) PRIME OR COMPOSITE:

```
num=int(input("Enter number: "))
for i in range(2,num//2+1):
    if(num%i==0):
        print("Not prime")
        break
else:
    print("Prime number")
```

OUTPUT:

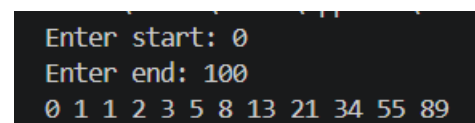


```
Enter number: 34
Not prime
PS C:\Users\tanwe\A
Enter number: 97
Prime number
```

4) FIBONACCI BETWEEN TWO NUMBERS:

```
start=int(input("Enter start: "))
end=int(input("Enter end: "))
a=0
b=1
while a<=end:
    if a>=start:
        print(a,end=" ")
    a,b=b,a+b
```

OUTPUT:



```
Enter start: 0
Enter end: 100
0 1 1 2 3 5 8 13 21 34 55 89
```

5) VOWEL, CONSONANT AND SPACE COUNT:

```
str=input("Enter string: ")
cvow=ccon=cspace=0
for i in str:
    if i in "aeiouAEIOU":
        cvow+=1
    elif i==" ":
        cspace+=1
    else:
        ccon+=1
print(f"Vowels: {cvow}, Consonants: {ccon}, Spaces: {cspace}")
```

OUTPUT:

```
Enter string: hello and welcome to EDA class one
Vowels: 12, Consonants: 16, Spaces: 6
```

6) LIST OPERATIONS:

```
lst=eval(input("Enter list: "))
print("Max = ",max(lst),"Min = ",min(lst))
print("Sum = ",sum(lst),"Reverse Order = ",lst[::-1])
```

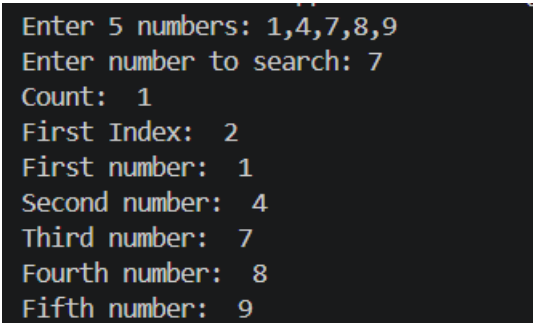
OUTPUT:

```
Enter list: [10,5,2,39,14,9]
Max = 39
Min = 2
Sum = 79
Reverse Order = [9, 14, 39, 2, 5, 10]
```

7) TUPLE FUNCTIONS AND UNPACKING

```
tup=eval(input("Enter 5 numbers: "))
num=int(input("Enter number to search: "))
print("Count: ",tup.count(num),"\nFirst Index: ",tup.index(num))
a,b,c,d,e=tup
print("First number: ",a)
print("Second number: ",b)
print("Third number: ",c)
print("Fourth number: ",d)
print("Fifth number: ",e)
```

OUTPUT:

A screenshot of a terminal window with a dark background and light-colored text. It shows the output of the Python code for the tuple functions. The input '1,4,7,8,9' is entered for the 5 numbers, and '7' is entered for the number to search. The output shows the count of '7' is 1 and its first index is 2. Then, each element of the tuple is printed in order: 1, 4, 7, 8, 9.

```
Enter 5 numbers: 1,4,7,8,9
Enter number to search: 7
Count: 1
First Index: 2
First number: 1
Second number: 4
Third number: 7
Fourth number: 8
Fifth number: 9
```

8) DICTIONARY INSERTION, SORTING AND SEARCHING

```
n=int(input("Enter number of students: "))
dict={}
for i in range(n):
    name=input("Enter name of student: ")
    marks=int(input("Enter marks of student: "))
    dict[name]=marks
srted=sorted(dict.items(),key=lambda x:x[1],reverse=True)
```

```
print("Highest Marks: ",srted[0][0]," with marks: ",srted[0][1])
name=input("Enter name of student to find marks: ")
print("Marks of ",name," : ",dict.get(name,"No marks found"))
```

OUTPUT:

```
Enter number of students: 3
Enter name of student: Jim
Enter marks of student: 92
Enter name of student: Tom
Enter marks of student: 100
Enter name of student: Klaus
Enter marks of student: 87
Highest Marks: Tom with marks: 100
Enter name of student to find marks: Klaus
Marks of Klaus : 87
```

9) WORD FREQUENCY

```
sentence=input("Enter a sentence: ")
words=sentence.split()
wcount = {}
for i in words:
    word = i.lower()
    if word in wcount:
        wcount[word]+=1
    else:
        wcount[word]=1
print("Word Frequencies:")
for word,count in wcount.items():
    print(word," : ",count)
```

OUTPUT:

```
Enter a sentence: hello hello this is EDA
Word Frequencies:
hello : 2
this : 1
is : 1
eda : 1
```

10) ROTATING LIST CLOCKWISE:

```
lst=[1,2,3,4,5]
n=int(input("Enter number of positions to rotate: "))
n=n%len(lst)
rotated=lst[-n:]+lst[:-n]
print("Rotated List:",rotated)
```

OUTPUT:

```
Enter number of positions to rotate: 3
Rotated List: [3, 4, 5, 1, 2]
```